USB PD CTS ENGINEERING CHANGE NOTICE

NOTICE: Any Company or Companies submitting a USB Power Delivery ECR proposal must be one of the following: a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum. If a group of Companies is submitting an ECR proposal, each company must be either a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum.

SPECIFICATION REVISIONS AND ADDENDA: At any point in time, there shall only be one current version of the USB PD CTS, termed the production version. At the same time, there may also be proposed revisions to the specification's design which are not yet approved and shall be held confidential as deemed necessary by the USB 3.0 and USB 2.0 Promoters and within the Group of Working Committee(s).

PROCEDURES FOR SUBMITTING PROPOSALS: Both members of the USB Implementers Forum as a whole and members of the USB 3.0 and USB 2.0 Promoters may submit requests to revise the USB PD CTS Specification. Such a request may be rejected or may result in a USB PD Engineering Change Notice (ECN), which is the official way USB specifications may be changed.

FORMAT OF PROPOSAL: The originator of a request to alter the USB PD CTS Specification may do so by posting this to the USB Power Delivery Compliance working group for review. Once the proposal has been reviewed by the working group it will be passed to the USB 3.0 and 2.0 Promoters for approval to publish.

RESUBMISSION AND APPEAL: The originator of a request that was not approved can redraft the original request. Rewritten proposal will be treated as a new proposal and will be evaluated using the procedures described above. The originator of a request that was not approved can also submit an appeal to the USB 3.0 and 2.0 Promoters. The appeal must be made in writing and addressed to the Secretary of the USB Implementers Forum.

ABOUT THE ENGINEERING CHANGE REQUEST FORM:
The Purpose of this Engineering Change Request Form is to expedite the review process of the proposal by providing explanations, background information, and examples of the proposed changes at a high level. This form serves as an executive summary to the actual proposal.

STEPS ON HOW TO SUBMIT A USB PD ENGINEERING CHANGE REQUEST:

1) Please fill out the Engineering Change Request Form on the following pages completely:
   a) Detail the names and contact details for each of the ECR contributors
   b) Update the ECR Title
   c) Give a minimum of 2-3 sentences for each description on the form outlining the background to the ECR
2) For each section/table/figure to be updated:
   a) Detail the section number, starting page and figure/table number to be updated as appropriate.
   b) Detail existing text under "From Text"
   c) Detail changed text under "To Text"
3) Save the file as "USB PD CTS 1.0 R 1" followed by the ECR Title as per step 1)b)
4) Post the ECR in the USB PD CTS Documents section under "ECR | New ECRs".
   a) This ECR will then be reviewed by the Power Delivery Compliance Working Group.
   b) Revisions to the ECR originating from the review should be submitted as document revision of the original ECR using "Add new document".

USB PD CTS ENGINEERING CHANGE NOTICE

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USB PD CTS ENGINEERING CHANGE NOTICE

Title: COMMON.CHECK.PD.2 Check Message Header Update
Applied to: USB PD CTS Q3, 2026

Brief description of the functional changes proposed:
Update COMMON.CHECK.PD - remove MessageIDCounter reset check after a PR Swap or FR Swap per approved ECR https://groups.usb.org/wg/powerdelivery/document/previewpdf/18925

Benefits as a result of the proposed changes:
Proper testing of the DUT designed per the updated MessageIDCounter reset rule.

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
None

An analysis of the hardware implications:
See https://groups.usb.org/wg/powerdelivery/document/previewpdf/18925

An analysis of the software implications:
See https://groups.usb.org/wg/powerdelivery/document/previewpdf/18925

An analysis of the compliance testing implications:
Tester to update the test per the ECR. Also do not issue a Soft Reset for DUTs not resetting MessageID in the Message header after a PR_Swap or FR_Swap

An analysis of the Vendor Info File (VIF) implications:
None

USB PD CTS ENGINEERING CHANGE NOTICE

Actual Change Requested

COMMON.CHECK.PD.2 Check Message Header

From Text:

COMMON.CHECK.PD.2 Check Message Header

Description: The Tester performs the protocol checks on the Message Header sent by the UUT.
Check Applicability: All PD messages except *GoodCRC.*
The Tester performs the following checks on the Message Header of all PD messages except *GoodCRC* and Returned BIST Counters: [COMMON.CHECK.PD.2#1]

1. Bit 7…6: (Specification Revision)
    a. For messages sent by the UUT as initial Source after detaching, *Hard Reset*, or *Error Recovery*, prior to Revision Alignment this check fails if this field is not set to 10b. This applies to:
        i. Messages sent over SOP prior to the first *Request* from the Tester and
        ii. Messages sent over SOP' prior to the first *Request* over SOP or the first *Discover Identity* response over SOP' from the Tester.
    b. For all other messages, the check fails if the Specification Revision differs from the revision specified by the present Test Condition.
2. Bits 14…12: Number of Data Objects
    a. When the Extended field is set to zero, the check fails if Number of Data Objects field does not match the number of 32-bit Data Objects that follow the Message Header
    b. When the Extended bit (in the Message Header) is set to 1b and Chunked bit (in the Extended Message Header) is set to 1b, the check fails if the size in bytes of the payload following the Message Header is not 4 times the Number of Data Objects
    c. When the Extended bit (in the Message Header) is set to 1b and Chunked bit (in the Extended Message Header) is set to 0b, the check fails if Number of Data Objects field is not set to 000b.
    d. Data messages:
        i. A *Request* Message: Number of Data Objects = 1
        ii. A *Battery_Status* Message: Number of Data Objects = 1
        iii. An *Alert* Message: Number of Data Objects = 1
        iv. A *Get_Country_Info* Message: Number of Data Objects = 1
        v. An *Enter_USB* Message: Number of Data Objects = 1
        vi. An *EPR_Request* Message: Number of Data Objects = 2
3. Bits 11…9: *MessageID*. The check fails if the value is not 000b under the following conditions:
    a. The first message on each SOP* type after a *Hard Reset* signaling was sent or received .
    b. The message is a *Soft_Reset* Message
    c. The first message on the same SOP* type after receiving *Soft_Reset*.
    d. The first message on SOP after a successful Power Role Swap
    e. The first message on SOP after a successful Fast Role Swap
    f. The first message on each SOP* type upon initial entry to Attached state.
    g. The first message on each SOP' and SOP" after a *Cable Reset* or *Data Reset*

To Text:

COMMON.CHECK.PD.2 Check Message Header

Description: The Tester performs the protocol checks on the Message Header sent by the UUT.
Check Applicability: All PD messages except *GoodCRC.*
The Tester performs the following checks on the Message Header of all PD messages except *GoodCRC* and Returned BIST Counters: [COMMON.CHECK.PD.2#1]

1. Bit 7…6: (Specification Revision)
    a. For messages sent by the UUT as initial Source after detaching, *Hard Reset*, or *Error Recovery*, prior to Revision Alignment this check fails if this field is not set to 10b. This applies to:
        i. Messages sent over SOP prior to the first *Request* from the Tester and
        ii. Messages sent over SOP' prior to the first *Request* over SOP or the first *Discover Identity* response over SOP' from the Tester.
    b. For all other messages, the check fails if the Specification Revision differs from the revision specified by the present Test Condition.
2. Bits 14…12: Number of Data Objects
    a. When the Extended field is set to zero, the check fails if Number of Data Objects field does not match the number of 32-bit Data Objects that follow the Message Header
    b. When the Extended bit (in the Message Header) is set to 1b and Chunked bit (in the Extended Message Header) is set to 1b, the check fails if the size in bytes of the payload following the Message Header is not 4 times the Number of Data Objects
    c. When the Extended bit (in the Message Header) is set to 1b and Chunked bit (in the Extended Message Header) is set to 0b, the check fails if Number of Data Objects field is not set to 000b.
    d. Data messages:
        i. A *Request* Message: Number of Data Objects = 1
        ii. A *Battery_Status* Message: Number of Data Objects = 1
        iii. An *Alert* Message: Number of Data Objects = 1
        iv. A *Get_Country_Info* Message: Number of Data Objects = 1
        v. An *Enter_USB* Message: Number of Data Objects = 1
        vi. An *EPR_Request* Message: Number of Data Objects = 2
        vii.
3. Bits 11…9: *MessageID*. The check fails if the value is not 000b under the following conditions:
    a. The first message on each SOP* type after a *Hard Reset* signaling was sent or received .
    b. The message is a *Soft_Reset* Message
    c. The first message on the same SOP* type after receiving *Soft_Reset*.
    ~~d. The first message on SOP after a successful Power Role Swap~~
    ~~e. The first message on SOP after a successful Fast Role Swap~~
    f. The first message on each SOP* type upon initial entry to Attached state.
    g. The first message on each SOP' and SOP" after a *Cable Reset* or *Data Reset*

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